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Solar ESS product sheet · October 2026

Solar panels

Back-contact panels from LONGi and AIKO

The newest step in a long line of improvements. Here is how panels got to this point.

Each generation of solar panel has turned a little more of the same sunlight into electricity.

1. **Amorphous and thin film**A thin layer of silicon or another material laid on glass or a flexible sheet. Cheap and light, and familiar from pocket calculators, but low in output, so a roof needed a large area for little power.
2. **Monocrystalline**Cells cut from a single silicon crystal. More power from the same area and a longer life, which made mono the standard for rooftops.
3. **PERC**A reflective layer added to the back of the cell sends unused light back through the silicon for a second chance at being absorbed.
4. **Bifacial**Glass on both sides, so the panel also collects light reflected from the ground or roof onto its back.
5. **Back contact (HPBC and ABC)**All the metal wiring moves to the back of the cell. Nothing on the front blocks the light, and the whole face of the cell works.

Our standard panel

LONGi HPBC

HPBC stands for Hybrid Passivated Back Contact, the technology in LONGi's Hi-MO X10 range. With no busbars on the front, more light reaches the cell and the panel has a clean, all-black face.

It gives high output per panel, so the same system needs fewer panels, fewer roof fixings and less cable. It also copes better when part of a panel is shaded by a tree, a pole or a neighbouring roof.

Also installed

AIKO ABC

ABC stands for All Back Contact, used in AIKO's Neostar range. Both the positive and negative contacts sit on the back of the cell, leaving the front completely clear.

It is among the most efficient panels available for rooftops, and handles partial shading well, which suits small or awkward roofs where every square metre counts.

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AIKO or LONGi?

On technology the two are almost a tie. Both lose 0.26% of power per °C, age at the same rate and carry the same 15-year product and 30-year performance warranties. In the same size frame, the AIKO sold in Thailand gives 670 W against 650 W for the LONGi.

On price there is no clear winner: in Thai shops both cost about the same per watt. LONGi is the larger company, shipping about three times as many panels. We choose between them by roof space and the supplier price on the day.

For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1. All qualifications and certificates will be furnished upon request or upon completion of contract.

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